Manohar Golleru

+1 716-398-6185 manohargolleru@outlook.com linkedin.com/in/manohar-golleru Portfolio

Education

- **Industrial and Systems Engineering, University at Buffalo, SUNY** Buffalo, NY
  Doctor of Philosophy (GPA: 3.94) *Aug 2024 – Expected May 2027*
  Concentration: Human Factors & Ergonomics
  Thesis: "Enhancing Engagement and Interaction in Augmentative and Alternative Communication (AAC)"
  Master of Science (GPA: 3.91) *May 2024*
- **Bachelor of Technology, Mechanical Engineering** *August 2021*
  Amrita Vishwa Vidyapeetham, India

Research & Analytical Experience

Communication and Assistive Device Lab (CADL), University at Buffalo Buffalo, NY
Research Assistant *Jan 2023 – Present*

- Built **retrieval-augmented generation (RAG) knowledge bases** from **PDF, text, and multimodal datasets** (transcripts, audio, session metadata) by extracting content, chunking, generating embeddings, and indexing in **FAISS** to support semantic search and retrieval.
- Architected a **RAG platform** using **LightRAG, GraphRAG, and FAISS vector search** to index conversational data and generate **real-time AI responses** for AAC users with speech and motor disabilities (ALS, cerebral palsy).
- Designed and deployed **DEAN (Dynamic Expressive Augmented Narrator)**, integrating **fine-tuned LLMs**, **advanced prompting**, Deepgram ASR, and **Microsoft Azure Neural Speech** to enable adaptive, expressive, low-latency conversational interaction.
- Built **backend AI services** using **FastAPI and Flask** to orchestrate **LLM inference**, **vector retrieval**, and speech pipelines; deployed **containerized services** on **Google Cloud Platform** with **Docker** and implemented monitoring via **Prometheus and Grafana**.
- Contributed to **OS-DPI** (Open-Source Developer Programming Interface), a **Node.js** front-end platform for dynamic AAC interface prototyping supporting real-time interaction and data collection.
- Conducted **user studies** on AAC-mediated communication combining **interface design**, **usability evaluation**, and **multimodal analysis (ELAN, R)** to assess engagement and drive iterative design improvements.
- Applied **human-centered design principles** and collaborated with interdisciplinary teams in **Communication Disorder Sciences** and **Computer Science** to advance AI-driven assistive technologies.

Surgery Ergonomics and Human Factors Lab (SurgE), University at Buffalo Buffalo, NY
Research Assistant *Apr 2024 – Present*

- Developed **machine learning pipelines** (tree-based models, SVM) on **high-frequency time-series biomarker data** from Empatica wearable sensors to detect alcohol impairment with approximately **80% classification accuracy**.
- Built **Python feature-engineering pipelines** to process large physiological datasets for model training, validation, and comparison.
- Conducted **statistical analyses** (**ANOVA, t-tests**) and created **data visualizations** to examine behavioral variation across study phases, supporting iterative model improvement.
- Designed a **web-based analytics application** capturing real-time **surgeon eye-tracking data** and generating **comparative heatmaps** for novice vs. expert gaze pattern evaluation.
- Planned and initiated a **human factors pilot study** evaluating **AI-assisted surgical systems**, defining measures for clinical decision-making, task performance, and patient safety.

Projects

- **Drone Show Simulation (PX4, Gazebo, ROS, MAVSDK)** *Jan 2025 – May 2025*
  - Built a multi-UAV formation control and flight testing platform using **PX4 SITL** integrated with **Gazebo** for realistic physics and **MAVSDK** for off-board command execution.
  - Implemented image-to-waypoint formation mapping using **OpenCV**.
- **mHealth App Interface Design (Figma)** *Aug 2022 – Dec 2022*
  - Developed low- and high-fidelity prototypes, conducted usability testing, and refined user workflows.

  - Secured **third place** at a university business pitch competition.

- **Workplace Ergonomic Analysis (RULA, ACGIH-TLV)** *2023*
  - Performed ergonomic risk assessments to identify musculoskeletal injury risks and implemented solutions improving workplace safety and efficiency.

Research Publications

- Pal, S., Murali, N., Jadhav, A. V., Bizovi, J., Satchidanand, A., **Golleru, M.**, Agarwal, S., Hutchinson, T., Higginbotham, J., & Srihari, R. K. (2026). *SPICA: Scalable and Personalized Conversational Agent Framework for AAC Users.* In *Proceedings of the 31st International Conference on Intelligent User Interfaces (IUI '26).* Association for Computing Machinery, New York, NY, USA, 218–235. https://doi.org/10.1145/3742413.3789116
- Quinn, T., Lopez, E., L'Huillier, J. C., Woodward, J. M., **Golleru, M.**, Clipp, R., Enquobahrie, A., Schwaitzberg, S. D., & Cavuoto, L. (2026). *Describing what surgeons see: Visual orientation strategies in laparoscopic cholecystectomy across imaging conditions. Journal of Surgical Education, 83*(7), 103968. `https://doi.org/10.1016/j.jsurg.2026.103968`
- Kaczor, E. E., **Golleru, M.**, Carter, D., Painter, O. B., Kelleran, K. J., Lynch, J. J., Clemency, B. M., Cavuoto, L., & Chai, P. R. (2026). *Detecting Ethanol Intoxication and Impairment Using Wearable Biosensors. Proceedings of the Hawaii International Conference on System Sciences*, 3532–3542.
- Higginbotham, J., **Golleru, M.**, Pal, S., Satchidanand, A., Bizovi, J., Hutchinson, T., Buckley, M., Mathy, P., Agarwal, S., & Srihari, R. K. (n.d.). *Developing Artificial Intelligence Applications for Human Conversation: Perspectives, Tools, Exploratory Analyses. Assistive Technology Outcomes and Benefits*, p. 13.
- **Golleru, M.** (2024). *Enhancing engagement and interaction in augmentative and alternative communication (AAC): A comparative study on the impact of front-facing partner display in SGD-mediated conversations.* ProQuest Dissertations & Theses.

Conference Presentations

- Kaczor, E. E., Golleru, M., Carter, D., Painter, O. B., Kelleran, K. J., Lynch, J. J., Clemency, B. M., Cavuoto, L., & Chai, P. R. (2026). *Detecting Ethanol Intoxication and Impairment Using Wearable Biosensors.* Platform presentation at the Hawaii International Conference on System Sciences, Maui, HI.
- Golleru, M., Higginbotham, J., Cavuoto, L., & Mathy, P. (2024, September 9–13). *Enhancing engagement and interaction in augmentative and alternative communication (AAC): A comparative study on the impact of front-facing partner display in SGD-mediated conversations* [Conference presentation]. ASPIRE HFES 68th International Annual Meeting, Phoenix, AZ.
- Higginbotham, J., Golleru, M., Bizovi, J., & Hutchinson, T. (2024, January 30–February 1). *Conversational artificial intelligence: Promises and current realities* [Education session]. Assistive Technology Industry Association (ATIA) Conference, Orlando, FL.

Leadership

- **President, HFES Student Chapter, University at Buffalo** *Feb 2026 – Present*
  Lead chapter operations, organize professional programming, and support student engagement in human factors.
- **Vice President, HFES Student Chapter, University at Buffalo** *Jan 2025 – Jan 2026*
  Supported events and partnerships; contributed to the chapter earning **Gold Level Recognition** (2024–2025).

Skills

- **Programming:** Python (pandas, NumPy), R, SQL, SAS, MATLAB, Node.js
- **Machine Learning / NLP:** PyTorch, TensorFlow, transformers, fine-tuning, prompt engineering, computer vision (OpenCV, YOLO, OpenPose, MediaPipe)
- **Data Engineering & Analytics:** RAG, FAISS vector search, multimodal ingestion, feature engineering, experiment design, time-series analysis, hypothesis testing, data visualization
- **Backend Development:** FastAPI, Flask, REST APIs, WebSockets
- **Cloud & DevOps:** Docker, Google Cloud Platform, Prometheus, Grafana
- **Simulation & Robotics:** ROS, PX4, Gazebo, MAVSDK
- **Human Factors Methods:** Usability testing, workflow analysis, mixed-methods research, ergonomic risk assessment (RULA, REBA, ACGIH-TLV)
- **Tools:** Git, ELAN, Figma

Certifications

- **CITI Program:** Conflict of Interest, Social & Behavioral Research, Responsible Conduct of Research
- **OSHA:** 501, 511, 7500, 7505